

prior b'' where $U_i(\mathcal{S}) = U_i(\mathcal{D})$ (in which case set $(b_i^*, b_i^{**}) = (b'', b'')$).

(iii) Otherwise, if $\tilde{b} < b'$, set $(b_i^*, b_i^{**}) = (\tilde{b}, b')$; however, if $\tilde{b} \geq b'$, then we set $(b_i^*, b_i^{**}) = (b', b')$.

This is of the cutoff form claimed in the lemma. ■

An immediate consequence of Lemma A.1 is that any Bayesian-Nash equilibrium must be in cutoff strategies for all agents. Hence, we can limit our attention to cutoff strategies (b_i^*, b_i^{**}) for every agent i , which can be represented as $(\mathbf{b}^*, \mathbf{b}^{**})$ in vector notation. This is a partially-ordered set according to the component-wise order $\succeq$. Hence, the cutoff space $\mathbf{B} = [0, 1]^{2N}$ forms a *complete lattice*.²⁵

Next, we define a map $\psi : \mathbf{B} \rightarrow \mathbf{B}$ that maps cutoffs $(\mathbf{b}^*, \mathbf{b}^{**})$ to best-response cutoffs $(\mathbf{b}^{*,BR}, \mathbf{b}^{**,BR})$. This map is well-defined because (i) H is a continuous distribution, so we need not specify the strategies of agents precisely on the cutoffs, and (ii) by Lemma A.1, for any set of strategies σ_{-i} (including the cutoff strategies given by $(\mathbf{b}^*, \mathbf{b}^{**})$), all agents' best responses are in cutoff form.

Lemma A.2. *The map ψ preserves the component-wise order $\succeq$.*

Proof of Lemma A.2. Consider some $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**}) \succeq (\mathbf{b}^*, \mathbf{b}^{**})$. Fixing an article with observables (r, m) , $U_i(\mathcal{D})$, $U_i(\mathcal{I})$ and $U_i^{(1)}$ are independent of $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$ and $(\mathbf{b}^*, \mathbf{b}^{**})$. However, for $U_i^{(2)}$ we have:

$$\begin{aligned} \mathbb{E}_{\mathbf{P}, (\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}[\kappa S_i - dD_i] &= \sum_{j=1}^N p_{ij} \left(\kappa \mathbb{P}_{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}[a_j = \mathcal{S}] - d \mathbb{P}_{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}[a_j = \mathcal{D}] \right) = \sum_{j=1}^N p_{ij} \left(\kappa \mathbb{P}_H[b_j > \hat{b}_j^{**}] - d \mathbb{P}_H[b_j < \hat{b}_j^*] \right) \\ &\leq \sum_{j=1}^N p_{ij} \left(\kappa \mathbb{P}_H[b_j > b_j^{**}] - d \mathbb{P}_H[b_j < b_j^*] \right) = \mathbb{E}_{\mathbf{P}, (\tilde{\mathbf{b}}^*, \tilde{\mathbf{b}}^{**})}[\kappa S_i - dD_i] \end{aligned}$$

As a result, $U_i^{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}(\mathcal{S}) \leq U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S})$. As in Lemma A.1, we define $\tilde{b}$ as the prior where $U_i(\mathcal{D}) = 0$ if such a $\tilde{b}$ exists, otherwise let $\tilde{b} = 0$ if ignoring dominates disliking and $\tilde{b} = 1$ if disliking dominates ignoring. Observe that $\tilde{b}$ is the same for both $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$ and $(\mathbf{b}^*, \mathbf{b}^{**})$. We have three cases for the best-response cutoffs $(\mathbf{b}^{*,BR}, \mathbf{b}^{**,BR})$ given other agents' cutoffs $(\mathbf{b}^*, \mathbf{b}^{**})$ (which we compare to $(\hat{\mathbf{b}}^{*,BR}, \hat{\mathbf{b}}^{**,BR})$ given other agents' cutoffs $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$):

- (i) Ignoring dominates sharing for agent i (for given cutoffs $(\mathbf{b}^*, \mathbf{b}^{**})$). Then by virtue of $U_i^{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}(\mathcal{S}) \leq U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S})$, ignoring dominates sharing with $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$ as well. Thus, $(b_i^{*,BR}, b_i^{**,BR}) = (\hat{b}_i^{*,BR}, \hat{b}_i^{**,BR}) = (\tilde{b}, 1)$.
- (ii) Sharing dominates ignoring for agent i (for given cutoffs $(\mathbf{b}^*, \mathbf{b}^{**})$). Then either sharing dominates disliking (in which case $(b_i^{*,BR}, b_i^{**,BR}) = (0, 0) \preceq (\hat{b}_i^{*,BR}, \hat{b}_i^{**,BR})$ trivially), or there exists some prior b'' where $U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S}) = U_i(\mathcal{D})$ denoted by b'' and $(b_i^{*,BR}, b_i^{**,BR}) = (b'', b'')$. Moreover, because $U_i^{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}(\mathcal{S}) \leq U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S}) = U_i(\mathcal{D})$ at prior b'' , for an agent with prior b'' , playing $\mathcal{D}$ is a (weakly) better response than sharing when other agents play according to cutoffs $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$. By monotonicity of $U_i(\mathcal{S})$ and $U_i(\mathcal{D})$ in prior b_i , this implies that $b_i^{**,BR} \leq \hat{b}_i^{**,BR}$. If ignoring is never a best response when $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$, then $\hat{b}_i^{*,BR} = \hat{b}_i^{**,BR}$. Otherwise, $\hat{b}_i^{*,BR} = \tilde{b} \geq b_i^{**,BR} = b_i^{*,BR}$.

²⁵Note that for any collection of cutoffs $\{(\mathbf{b}^{*,(1)}, \mathbf{b}^{**, (1)}), (\mathbf{b}^{*,(2)}, \mathbf{b}^{**, (2)}), \dots, \}$ in the cutoff space, there is a greatest lower bound given by the component-wise infimum and a least upper bound given by the component-wise supremum.